1.4 The Euler–Lagrange Equation for Local Field

In the preceding sections, we varied the particle's coordinates, as demonstrated in Eq. (1.2.2), in order to derive the equations of motion, that is, to determine the particle's trajectory. When we shift our focus from particles to fields themselves, however, the equations we seek are no longer equations of motion but **field equations**, and the quantities with respect to which we perform variations must change accordingly.

In this case, we need to vary the quantities that describe the fields. For example, when studying the electromagnetic field, we vary the electromagnetic potentials. When studying the gravitational field, which we interpret as the curvature of spacetime, we vary the metric that describes the geometry of spacetime. When studying quantum fields, we represent them in terms of field operators and therefore vary the field operators themselves. In this section, we set aside the specific physical systems under consideration and derive the Euler–Lagrange equations for local fields in a general form.

Let us denote a field by φ_a. This field may be a scalar field, a vector field, or a spinor field; the subscript a simply labels the chosen field components. The Lagrangian of the field can then be expressed as a scalar function of the field and its spacetime derivatives:

$$L = L(\varphi_a, \partial_i \varphi_a) \tag{1.4.1}$$

Since the action of a field is a scalar, the expression in Eq. (1.4.1) must be integrated over a scalar measure. Following Eq. (1.3.13), we integrate Eq. (1.4.1) over the spacetime line element ds, obtaining

$$S = \int L(\varphi_a, \partial_i \varphi_a)\, ds = \int L(\varphi_a, \partial_i \varphi_a)\, c d\tau \tag{1.4.2}$$

We now introduce a new functional $\mathcal{L}(\varphi_a, \partial_i \varphi_a)$, defined such that

$$\int \mathcal{L}(\varphi_a, \partial_i \varphi_a)\, dV = L(\varphi_a, \partial_i \varphi_a) \tag{1.4.3}$$

Substituting $\mathcal{L}(\varphi_a, \partial_i \varphi_a)$ into Eq. (1.4.2), we obtain

$$S = \int \int \mathcal{L}(\varphi_a, \partial_i \varphi_a)\, c dV d\tau \tag{1.4.4}$$

Using Eq. (1.3.4), we can further write

$$S = \int \mathcal{L}(\varphi_a, \partial_i \varphi_a) \frac{dV}{\sqrt{1-\left(\frac{v}{c}\right)^2}} c \cdot \sqrt{1-\left(\frac{v}{c}\right)^2}\, dt = \int \mathcal{L}(\varphi_a, \partial_i \varphi_a) c dV dt$$
$$= \int \mathcal{L}(\varphi_a, \partial_i \varphi_a) d\Omega \tag{1.4.5}$$

Here, $d\Omega$ denotes the four-dimensional spacetime volume element. We can now see that $\mathcal{L}(\varphi_a, \partial_i \varphi_a)$ represents the Lagrangian density in four-dimensional spacetime, which we refer to as the **Lagrangian density**.

From the derivation of Eq. (1.4.5), we can see that this four-dimensional volume element is an invariant. This is because our present discussion concerns physical phenomena in flat spacetime. In Chapter 3, we will need to introduce an additional scalar quantity involving the metric in order to construct an invariant integration measure. Returning to Eq. (1.4.5), we take the variation of both sides and obtain

$$\begin{aligned} \delta S &= \int \left[\frac{\partial \mathcal{L}}{\partial \varphi_a} \delta \varphi_a + \frac{\partial \mathcal{L}}{\partial(\partial_i \varphi_a)} \delta(\partial_i \varphi_a)\right] d\Omega \\ &= \int \left[\frac{\partial \mathcal{L}}{\partial \varphi_a} \delta \varphi_a + \frac{\partial \mathcal{L}}{\partial(\partial_i \varphi_a)} \partial_i(\delta \varphi_a)\right] d\Omega \\ &= \int \left\{\frac{\partial \mathcal{L}}{\partial \varphi_a} \delta \varphi_a + \partial_i \left[\frac{\partial \mathcal{L}}{\partial(\partial_i \varphi_a)} \cdot \delta \varphi_a\right] - \partial_i \left[\frac{\partial \mathcal{L}}{\partial(\partial_i \varphi_a)}\right] \right. \\ &\quad \left. \cdot \delta \varphi_a\right\} d\Omega \end{aligned} \tag{1.4.6}$$

The middle term in the last line corresponds to the divergence term in the four-dimensional version of the Gauss theorem. Since the variation vanishes at spatial and temporal infinity, the integral of this term over the entire spacetime region is zero. Therefore, Eq. (1.4.6) reduces to

$$\delta S = \int \left\{\frac{\partial \mathcal{L}}{\partial \varphi_a} - \partial_i \left[\frac{\partial \mathcal{L}}{\partial(\partial_i \varphi_a)}\right]\right\} \cdot \delta \varphi_a d\Omega = 0 \tag{1.4.7}$$

Since the variation $\delta \varphi_a$ is arbitrary, the condition in Eq. (1.4.7) can hold only if

$$\frac{\partial \mathcal{L}}{\partial \varphi_a} - \partial_i \left[\frac{\partial \mathcal{L}}{\partial(\partial_i \varphi_a)}\right] = 0 \tag{1.4.8}$$

This is the **Euler–Lagrange equation for local fields**. With Eq. (1.4.8), we can conveniently derive the field equations for electromagnetism, gravitation, and, subsequently, quantum fields. Quantum fields, however, possess certain special properties that require us to introduce group theory and concepts related to continuous media. We will systematically discuss these topics in Chapter 4.